☐ Modes of connection: ○ Wired • Ethernet ○ Wireless • Wi-Fi • Bluetooth ☐ Encryption ☐ IP addressing and MAC addressing ☐ Standards ☐ Common protocols including: ○ TCP/IP (Transmission Control Protocol/Internet Protocol) ○ HTTP (Hyper Text Transfer Protocol) ○ HTTPS (Hyper Text Transfer Protocol Secure) ○ FTP (File Transfer Protocol) ○ POP (Post Office Protocol) ○ IMAP (Internet Message Access Protocol) ○ SMTP (Simple Mail Transfer Protocol) ☐ The concept of layers	**Required** ✓ Compare benefits and drawbacks of wired versus wireless connection ✓ Recommend one or more connections for a given scenario ✓ The principle of encryption to secure data across network connections ✓ IP addressing and the format of an IP address (IPv4 and IPv6) ✓ A MAC address is assigned to devices; its use within a network ; its format ✓ The principle of a standard to provide rules for areas of computing ✓ Standards allows hardware/software to interact across different manufacturers/producers ✓ The principle of a (communication) protocol as a set of rules for transferring data ✓ That different types of protocols are used for different purposes ✓ The basic principles of each protocol i.e. its purpose and key features ✓ How layers are used in protocols, and the benefits of using layers; for a teaching example, please refer to the 4-layer TCP/IP model **Not required** ✗ Understand how Ethernet, Wi-Fi and Bluetooth protocols work ✗ Understand differences between static and dynamic, or public and private IP addresses ✗ Knowledge of individual standards ✗ Knowledge of the names and function of each TCP/IP layer